Coefficient-Cancellation Templates and Polynomial Families of Fourth-Power Identities

Derivations, Specializations, and a Complementary Cancellation Pattern

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Abstract

This paper develops explicit polynomial identities involving fourth powers and a variable polynomial coefficient. The main tool is the factorization

$$(X+Y)^4-(X-Y)^4=8XY(X^2+Y^2).$$

Two previously stated coefficient-cancellation templates are proved and their motivating identities recovered as special cases. A third, complementary template is then developed by expanding the same symmetric fourth-power difference in five monomials. We determine all nonzero rational parameter choices for which the three middle coefficients vanish, producing a sparse two-endpoint residual family that contains a specified identity in $\mathbb{Z}[x,y]$. A separate binomial-expansion family provides further examples. The results establish algebraic identities and parameter conditions; they do not claim a classification of quartic Diophantine equations or a priority result.

Keywords: fourth-power identities; polynomial identities; coefficient cancellation; quartic Diophantine equations.
MSC 2020: 11D25, 11D41.

1 Introduction and starting identities

The purpose of this paper is to organize several constructions of polynomial identities between fourth powers and to give a complementary coefficient-cancellation family. Let p, q denote indeterminates. The first motivating identity is

$$\begin{aligned}&(20p^8-q^8)(2q^7)^4+(64p^8q-2q^9)^4\\&\quad+(64p^9-8p^5q^4-3pq^8)^4=(64p^9+8p^5q^4-3pq^8)^4.\end{aligned}\tag{1}$$

A version of this identity appears in the public discussion cited in [1,2].

The second motivating identity is

$$\begin{aligned}&-2(190p^8+243q^8)(16p^7q^3)^4+\left[4p^3q(16p^8+27q^8)\right]^4\\&\quad+(32p^{12}-64p^8q^4-81q^{12})^4=(32p^{12}+64p^8q^4+81q^{12})^4.\end{aligned}\tag{2}$$

Its algebraic derivation is included in Section 4. The new construction in Section 5 concerns the identity

$$\begin{aligned}&(8x^{32}-729y^{32})(x^3y)^4+(2x^{11}y+9x^3y^9)^4\\&\quad+(x^{12}-3x^8y^4-9y^{12})^4=(x^{12}+3x^8y^4+9y^{12})^4.\end{aligned}\tag{3}$$

We prove that this identity comes from a five-coefficient expansion and characterize the nonzero rational parameters for which its three intermediate coefficients cancel.

Throughout, an equality in $\mathbb{Q}[p,q]$ is a polynomial identity and therefore remains valid after every integer specialization of its variables. A polynomial identity with a varying coefficient is not, by itself, a result that every fixed integer has infinitely many representations by fourth powers.

2 A shared factorization

For commuting quantities X, Y, direct expansion gives

$$(X+Y)^4-(X-Y)^4=8XY(X^2+Y^2)=8X^3Y+8XY^3. \tag{4}$$

The first two templates use this difference and subtract a further fourth power to cancel selected coefficients. The complementary construction in Section 5 uses the same factorization but leaves the two endpoint coefficients instead of the two middle residual coefficients that appear in Theorem 4.1.

3 The first coefficient-cancellation family

Theorem 3.1 (First general template)**.** *Let $A, G, R, S, T, K \in \mathbb{Q}$ with $AK \neq 0$, and suppose*

$$R^4=8GA^3, \qquad T=-\frac{3G^2}{A}, \qquad S=-\frac{2RG^2}{A^2}.$$

Define

$$N_1(p,q)=\frac{40G^7}{A^3K^4}p^8-\frac{128G^9}{A^5K^4}q^8.$$

Then

$$\begin{aligned} &N_1(p,q)(Kq^7)^4+[q(Rp^8+Sq^8)]^4 \\ &\quad+[p(Ap^8+Tq^8-Gp^4q^4)]^4=[p(Ap^8+Tq^8+Gp^4q^4)]^4. \end{aligned} \tag{5}$$

Proof. Put $X=p(Ap^8+Tq^8)$, $Y=Gp^5q^4$, and $B=q(Rp^8+Sq^8)$. The last two fourth powers in (5) are $(X-Y)^4$ and $(X+Y)^4$. By (4),

$$\begin{aligned} (X+Y)^4-(X-Y)^4=8GA^3p^{32}q^4+(24GA^2T+8G^3A)p^{24}q^{12} \\ +(24GAT^2+8G^3T)p^{16}q^{20}+8GT^3p^8q^{28}. \end{aligned}$$

Meanwhile,

$$B^4=R^4p^{32}q^4+4R^3Sp^{24}q^{12}+6R^2S^2p^{16}q^{20}+4RS^3p^8q^{28}+S^4q^{36}.$$

The conditions on R, S, T cancel the coefficients of $p^{32}q^4$, $p^{24}q^{12}$, and $p^{16}q^{20}$ in the difference $(X+Y)^4-(X-Y)^4-B^4$. The remainder is

$$\frac{40G^7}{A^3}p^8q^{28}-\frac{128G^9}{A^5}q^{36}=N_1(p,q)(Kq^7)^4,$$

which proves the identity. □

3.1 Recovery of the first motivating identity

Take

$$(A, G, R, S, T, K) = (64, 8, 64, -2, -3, 2).$$

These values satisfy the hypotheses and give $N_1(p, q) = 20p^8 - q^8$. The middle base is $q(64p^8 - 2q^8) = 64p^8q - 2q^9$, and the symmetric pair is

$$p(64p^8 - 3q^8 \mp 8p^4q^4) = 64p^9 - 3pq^8 \mp 8p^5q^4.$$

Thus Theorem 3.1 yields (1).

3.2 A further specialization

The choice

$$(A, G, R, S, T, K) = (4, 8, 8, -64, -48, 2)$$

also satisfies the hypotheses. It gives

$$\begin{aligned}(81920p^8 - 1048576q^8)(2q^7)^4 + (8p^8q - 64q^9)^4 \\ + (4p^9 - 8p^5q^4 - 48pq^8)^4 = (4p^9 + 8p^5q^4 - 48pq^8)^4.\end{aligned}$$

This is a valid specialization of the template. Its status relative to the literature has not been determined, so no novelty claim is made.

4 The second coefficient-cancellation family

Theorem 4.1 (Second general template)**.** *Let $U, V, W, H, J, L \in \mathbb{Q}$ with $L \neq 0$, and suppose*

$$H^4 = 8U^3V, \qquad J^4 = 8UW^3, \qquad HJ^3 = 6UVW^2.$$

Define

$$N_2(p, q) = \frac{8U^3W + 8UV^3 - 4H^3J}{L^4}p^8 + \frac{24UV^2W - 6H^2J^2}{L^4}q^8.$$

Then

$$\begin{aligned}N_2(p, q)(Lp^7q^3)^4 + [p^3q(Hp^8 + Jq^8)]^4 \\ + [Up^{12} - q^4(Vp^8 + Wq^8)]^4 = [Up^{12} + q^4(Vp^8 + Wq^8)]^4.\end{aligned} \tag{6}$$

Proof. Set

$$X = Up^{12}, \qquad Y = q^4(Vp^8 + Wq^8), \qquad B = p^3q(Hp^8 + Jq^8).$$

Using (4) gives

$$\begin{aligned}(X + Y)^4 - (X - Y)^4 = 8U^3Vp^{44}q^4 + (8U^3W + 8UV^3)p^{36}q^{12} \\ + 24UV^2Wp^{28}q^{20} + 24UVW^2p^{20}q^{28} \\ + 8UW^3p^{12}q^{36}.\end{aligned}$$

Also,

$$\begin{aligned}B^4 = H^4p^{44}q^4 + 4H^3Jp^{36}q^{12} + 6H^2J^2p^{28}q^{20} \\ + 4HJ^3p^{20}q^{28} + J^4p^{12}q^{36}.\end{aligned}$$

The three assumptions cancel the coefficients of $p^{44}q^4$, $p^{20}q^{28}$, and $p^{12}q^{36}$ in $(X+Y)^4-(X-Y)^4-B^4$. The remaining terms are

$$\begin{aligned}(8U^3W + 8UV^3 - 4H^3J)p^{36}q^{12} \\ + (24UV^2W - 6H^2J^2)p^{28}q^{20} = N_2(p, q)(Lp^7q^3)^4,\end{aligned}$$

which proves the theorem. □

4.1 Recovery of the second motivating identity

Choose

$$(U, V, W, H, J, L) = (32, 64, 81, 64, 108, 16).$$

The required conditions hold, and

$$N_2(p, q) = -380p^8 - 486q^8 = -2(190p^8 + 243q^8).$$

Moreover,

$$B = 4p^3q(16p^8 + 27q^8), \qquad Y = q^4(64p^8 + 81q^8).$$

The symmetric bases are

$$32p^{12} \mp (64p^8q^4 + 81q^{12}),$$

so Theorem 4.1 gives exactly (2). This is a polynomial identity, not merely a numerical specialization.

5 A complementary coefficient-cancellation family

The identity (3) suggests a different cancellation pattern within the same expansion used in Theorem 4.1. We first state the residual identity for arbitrary parameters, then characterize the parameters that make the three middle coefficients vanish.

5.1 The five-coefficient identity

Let $U, V, W, H, J \in \mathbb{Q}$, and put

$$X = Up^{12}, \qquad Y = q^4(Vp^8 + Wq^8), \qquad B = p^3q(Hp^8 + Jq^8).$$

Define

$$\begin{aligned} c_0 &= 8U^3V - H^4, \\ c_1 &= 8U^3W + 8UV^3 - 4H^3J, \\ c_2 &= 24UV^2W - 6H^2J^2, \\ c_3 &= 24UVW^2 - 4HJ^3, \\ c_4 &= 8UW^3 - J^4, \end{aligned}$$

and let

$$C(p, q) = c_0p^{32} + c_1p^{24}q^8 + c_2p^{16}q^{16} + c_3p^8q^{24} + c_4q^{32}. \tag{7}$$

Theorem 5.1 (General five-coefficient identity)**.** *For all $U, V, W, H, J \in \mathbb{Q}$,*

$$C(p, q)(p^3q)^4 + B^4 + (X - Y)^4 = (X + Y)^4 \tag{8}$$

in $\mathbb{Q}[p, q]$.

Proof. By (4),

$$(X + Y)^4 - (X - Y)^4 = 8X^3Y + 8XY^3.$$

Expanding this expression and subtracting

$$B^4 = H^4p^{44}q^4 + 4H^3Jp^{36}q^{12} + 6H^2J^2p^{28}q^{20} + 4HJ^3p^{20}q^{28} + J^4p^{12}q^{36}$$

gives

$$\begin{aligned} &(X + Y)^4 - (X - Y)^4 - B^4 \\ &\quad = c_0p^{44}q^4 + c_1p^{36}q^{12} + c_2p^{28}q^{20} + c_3p^{20}q^{28} + c_4p^{12}q^{36} \\ &\quad = C(p, q)(p^3q)^4. \end{aligned}$$

Rearranging proves (8). □

5.2 Cancellation of the three middle coefficients

Theorem 5.2 (Sparse complementary family)**.** *Suppose $U, V, W, H, J \in \mathbb{Q} \setminus \{0\}$. Then $c_1 = c_2 = c_3 = 0$ if and only if there exist nonzero $u, t \in \mathbb{Q}$ such that*

$$U = u, \quad V = 3ut^4, \quad W = 9ut^{12}, \quad H = 2ut, \quad J = 9ut^9. \tag{9}$$

For these parameters, $c_0 = 8u^4t^4$ and $c_4 = -729u^4t^{36}$. Consequently, for all rational u, t,

$$\begin{aligned} &u^4(8t^4p^{32} - 729t^{36}q^{32})(p^3q)^4 \\ &\quad + \left(up^3q(2tp^8 + 9t^9q^8)\right)^4 \\ &\quad + \left(u(p^{12} - 3t^4p^8q^4 - 9t^{12}q^{12})\right)^4 \\ &= \left(u(p^{12} + 3t^4p^8q^4 + 9t^{12}q^{12})\right)^4. \end{aligned} \tag{10}$$

Proof. The equations $c_1 = c_2 = c_3 = 0$ are equivalent to

$$H^3J = 2U^3W + 2UV^3, \tag{11}$$

$$H^2J^2 = 4UV^2W, \tag{12}$$

$$HJ^3 = 6UVW^2. \tag{13}$$

Since the parameters are nonzero, dividing (13) by (12) gives

$$\frac{J}{H} = \frac{3W}{2V}.$$

Substituting this relation into (12) yields

$$H^4 = \frac{16UV^4}{9W}.$$

Using these equations in (11) gives $V^3 = 3U^2W$. Hence

$$H^4 = \frac{16}{3}U^3V.$$

Set $u = U$ and $t = H/(2U)$. It follows that $V = 3ut^4$, then $W = V^3/(3U^2) = 9ut^{12}$, and finally $J = 9ut^9$. This proves the forward implication. Direct substitution verifies the converse.

The endpoint coefficients are

$$\begin{aligned} c_0 &= 8u^3(3ut^4) - (2ut)^4 = 8u^4t^4, \\ c_4 &= 8u(9ut^{12})^3 - (9ut^9)^4 = -729u^4t^{36}. \end{aligned}$$

Substituting these values and (9) into Theorem 5.1 proves (10). The resulting identity itself remains valid when $u = 0$ or $t = 0$, by direct substitution, even though the parameter classification assumes nonzero values. □

5.3 The motivating identity and its relation to Theorem 4.1

Taking $u = t = 1$, $p = x$, and $q = y$ in (10) yields exactly (3). In the five-coefficient notation, the parameter choice is

$$(U, V, W, H, J) = (1, 3, 9, 2, 9).$$

The three middle coefficients vanish:

$$c_1 = 72 + 216 - 288 = 0, \quad c_2 = 1944 - 1944 = 0, \quad c_3 = 5832 - 5832 = 0,$$

whereas the endpoint coefficients are $c_0 = 24 - 16 = 8$ and $c_4 = 5832 - 6561 = -729$.

The relation to Theorem 4.1 is especially transparent in this notation. Its hypotheses are precisely $c_0 = c_3 = c_4 = 0$, leaving the two residual middle terms c_1, c_2. The new sparse family instead imposes $c_1 = c_2 = c_3 = 0$, leaving the endpoint terms c_0, c_4. Thus it is not a direct specialization of Theorem 4.1's stated parameter conditions. Theorem 3.1 uses a different arrangement of the symmetric pair and the auxiliary fourth power, so the identity is not presented as a direct specialization of that template either.

Remark 5.3. The family (10) is obtained from (3) by the substitution $x = p$, $y = tq$, followed by multiplying each fourth-power base by u (equivalently, multiplying the full identity by u^4). Thus the family is a structural generalization under variable substitution and scaling, not an essentially new independent family of quartic solutions.

6 A binomial-expansion family

The following independent construction follows directly from the binomial theorem. It is useful as a generator of examples, while its elementary origin should be kept in view.

Theorem 6.1 (Binomial family)**.** *For integers λ, μ, r, define*

$$P_{\lambda,\mu,r}(p,q) = 4(\mu-\lambda)p^{12} + 6(\mu^2-\lambda^2)p^8q^4 + 4(\mu^3-\lambda^3)p^4q^8 + (\mu^4-\lambda^4-r^4)q^{12}.$$

Then

$$P_{\lambda,\mu,r}(p,q)q^4 + (rq^4)^4 + (p^4+\lambda q^4)^4 = (p^4+\mu q^4)^4. \tag{14}$$

Proof. For an indeterminate s,

$$(p^4+sq^4)^4 = p^{16} + 4sp^{12}q^4 + 6s^2p^8q^8 + 4s^3p^4q^{12} + s^4q^{16}.$$

Subtract the expansion with $s = \lambda$ from the one with $s = \mu$, and then subtract $(rq^4)^4 = r^4q^{16}$. The result is exactly $P_{\lambda,\mu,r}(p,q)q^4$, proving (14). □

6.1 Three specializations

For $(\lambda, \mu, r) = (-4, 2, 6)$,

$$P_{-4,2,6} = 24(p^4-4q^4)(p^8+p^4q^4+16q^8),$$

so

$$24(p^4-4q^4)(p^8+p^4q^4+16q^8)q^4 + (6q^4)^4 + (p^4-4q^4)^4 = (p^4+2q^4)^4.$$

This specialization appears in the public discussion [2], where it is related to a rescaled Gerardin-type identity.

For $(\lambda, \mu, r) = (1, 3, 2)$,

$$P_{1,3,2} = 8p^{12} + 48p^8q^4 + 104p^4q^8 + 64q^{12},$$

and hence

$$(8p^{12} + 48p^8q^4 + 104p^4q^8 + 64q^{12})q^4 + (2q^4)^4 + (p^4+q^4)^4 = (p^4+3q^4)^4.$$

For $(\lambda, \mu, r) = (-2, 1, 3)$,

$$P_{-2,1,3} = 12p^{12} - 18p^8q^4 + 36p^4q^8 - 96q^{12},$$

which gives

$$\begin{aligned}(12p^{12} - 18p^8q^4 + 36p^4q^8 - 96q^{12})q^4 + (3q^4)^4 \\ + (p^4 - 2q^4)^4 = (p^4 + q^4)^4.\end{aligned}$$

7 Discussion and concluding remarks

The constructions above use two related but distinct ideas. The first two templates in Sections 3 and 4 select auxiliary bases so that specified coefficients in the symmetric fourth-power difference cancel. The complementary family in Section 5 uses the same five-term expansion as Theorem 4.1, but reverses which coefficients remain: the first construction leaves two middle coefficients, while the complementary construction leaves the two endpoint coefficients. The binomial family in Section 6 is a separate and more elementary source of identities.

The statements establish exact polynomial equalities and, in Theorem 5.2, an if-and-only-if description of the nonzero rational parameters that cancel the three middle coefficients in the specified five-coefficient setup. They do not give a complete classification of all fourth-power identities, prove that a fixed integer has infinitely many representations for a fixed coefficient, or settle primitivity and positivity of integer solutions. No claim of novelty or historical priority is made for the displayed identities.

Possible further work includes classifying the templates under scaling and variable substitution, identifying when different parameter choices yield equivalent identities, and separately studying whether particular specializations lead to infinitely many solutions of fixed-coefficient quartic equations.

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